ECEN 4/5632 Theory and Application of Digital Filters Fall 2026

Exit Ticket Questions for Week 6

1. A block has length B and the FIR has length L_h. Explain why the overlap-add method to implement linear convolution using the DFT/FFT requires

$$N_{\text{FFT}} \geq B + L_h - 1$$

and describe exactly what happens if the FFT is shorter.

2. You need to detect when a short transient occurs and also estimate its dominant frequency accurately. Explain what happens as you make the STFT window shorter or longer.

Good answers:

1. A strong answer should say that the block's linear convolution has length

$$B + L_h - 1,$$

that DFT multiplication computes circular convolution, and that insufficient FFT length causes the convolution tail to wrap onto the beginning and corrupt the result.

2. A shorter window improves time localization but broadens spectral features and worsens frequency localization. A longer window improves frequency localization but smears the transient over time. There is no window that simultaneously gives arbitrarily good resolution in both domains.

A good exit ticket answer should be short, correct, mathematically sound, to the point and demonstrate a well-reasoned understanding of the subject that the question is asking about. Exit ticket answers will be graded on a scale of “low” (1 point), “medium” (2 points), or “high” (3 points).

Use LaTeX style for math expressions, e.g., \delta[n] for a discrete-time impulse, or use the math equations in the rich content editor in Canvas.